Latto-Latto Control Research Changelog

Last updated: 2026-05-22

This document records major changes to the Latto-latto model, reward design, and training setup.

Rule for future updates. Any major change to the model, reward, controller, training setup, or evaluation protocol should be added to this file.

2026-05-22

Change 1: Reward Updated to Penalize Vertical Drift

Motivation.

- The original sparse reward was able to produce bouncing behavior, but the pivot vertical position z could drift over time.
- This made the learned behavior less physically meaningful and less suitable for fair controller comparison.

Previous reward.

$$r_t = \begin{cases} 1, & \text{if a collision occurs and the collision side alternates,} \\ 0, & \text{otherwise.} \end{cases}$$

Updated reward.

$$r_t = r_t^{\text{collision}} - \alpha z_t^2$$

with

$$r_t^{\text{collision}} = \begin{cases} 1, & \text{if a collision occurs and the collision side alternates,} \\ 0, & \text{otherwise.} \end{cases}$$

where:

- z_t is the vertical position of the pivot.
- α is the vertical position penalty weight.

Current setting in code.

$$\alpha = 0.1$$

Implementation notes.

- The reward is implemented in `latto_latto_model.py`.
- The environment now exposes reward components through the `info` dictionary:
  - `sparse_reward`
  - `z_position_penalty`

Observed outcome.

- PPO training completed successfully with the modified reward.
- The learned policy strongly reduced vertical drift.
- In quick evaluation, the policy also collapsed to a near-stationary solution with zero alternating collisions.

Interpretation.

- The drift penalty successfully regularized the base motion.
- The penalty weight $\alpha = 0.1$ appears too strong relative to the sparse collision reward.
- A smaller penalty weight, such as 0.01 or 0.02, should be tested next.

Change 2: Separate PPO Checkpoint for Reward-Penalty Experiment

Motivation. Preserve the original PPO checkpoint while training a new policy with the modified reward.

Change. Training output path changed from `ppo_latto` to `ppo_latto_z_penalty`.

Result. The reward-penalty experiment can now be evaluated independently from the original sparse-reward PPO model.

Change 3: Testing Script Updated to Load the New Checkpoint

Motivation. Keep the default evaluation script aligned with the most recent model variant.

Change. `testing_latto.py` now loads `ppo_latto_z_penalty`.

Note. If multiple controller or reward variants are added later, this script should eventually accept a model path as an argument instead of hardcoding one checkpoint.

Change 4: Visualization Workflow Improved

Motivation. The rendering logic was previously tied to repeated global `matplotlib` calls, which was less stable for repeated runs.

Change.

- Rendering now uses persistent `matplotlib` figure and axes handles.
- The environment now includes a `close()` method to release plotting resources cleanly.
- `testing_latto.py` now handles interactive and non-interactive backends more safely.

Research relevance. This is not a dynamics change, but it improves repeatability of visual inspection during controller evaluation.

Change 5: Testing Script Now Plots Pose Trajectories

Motivation.

- The previous testing workflow only saved the control input history.
- For controller comparison, it is also important to inspect how the pivot position and pendulum angle evolve over time.

Change.

- `testing_latto.py` now records and plots:
  - pivot vertical position $z(t)$
  - pendulum angle $\theta(t)$
- The script now saves an additional figure, `pose.png`.

Research relevance. This improves the evaluation workflow by making drift, oscillation amplitude, and near-stationary collapse easier to detect visually.